

Let the following equation represent a Periodic Autoregressive (PAR) model of order p , which we will call PAR(p):

$$Z_t = \sum_{i=1}^p \phi_i Z_{t-i} + \varepsilon_t$$

where:

- Z_t is the value of the time series at time t ,
- ϕ_i are the autoregressive coefficients for lag i ,
- ε_t is a white noise error term at time t .

We call Z our time series variable, but actually, for our purposes, it is the residual of an already de-seasonalized observational variable Y .

For this particular example, we consider Y to be a monthly inflow time series, de-seasonalized by simply normalizing each month by its mean and standard deviation. So, for our case,

$$Z_t = \frac{Y_t - \mu_{m(t)}}{\sigma_{m(t)}}$$

where:

- $\mu_{m(t)}$ is the mean of month m corresponding to time t ,
- $\sigma_{m(t)}$ is the standard deviation of month m corresponding to time t .

The full equation of the PAR(p) model for our de-seasonalized inflow time series Y can thus be expressed as:

$$\frac{Y_t - \mu_{m(t)}}{\sigma_{m(t)}} = \sum_{i=1}^p \phi_i \left(\frac{Y_{t-i} - \mu_{m(t-i)}}{\sigma_{m(t-i)}} \right) + \varepsilon_t$$

Our current approach for inserting the PAR coefficients into the Linear Programming (LP) model involves representing two variables inside the problem: the actual value of Y_t and its corresponding de-seasonalized value Z_t . The LP model will then include constraints that enforce the relationship defined by the PAR(p) model.

The first constraint is the actual AR dynamics, whose coefficients of lagged inflows are the PAR coefficients ϕ_i . These coefficients multiply the lagged inflows' de-seasonalized values Z_{t-i} , which are also represented as variables in the LP model.

The second constraint convert the de-seasonalized values Z_t back to the actual inflow values Y_t using the mean and standard deviation for the corresponding month. However, we think that this approach is not very efficient, and we like to not represent the de-seasonalized values Z_t as variables in the LP model. Instead, we would like to directly insert the PAR dynamics into the LP model using only the actual inflow values Y_t .

In the following, we will derive a conversion between the PAR coefficients ϕ_i and new coefficients ψ_i that can be used directly with the actual inflow values Y_t in the LP model. This will also give us a new noise term η_t that will replace the original white noise term ε_t .

Starting from the PAR(p) equation for Z_t :

$$\frac{Y_t - \mu_{m(t)}}{\sigma_{m(t)}} = \sum_{i=1}^p \phi_i \left(\frac{Y_{t-i} - \mu_{m(t-i)}}{\sigma_{m(t-i)}} \right) + \varepsilon_t$$

We can multiply both sides by $\sigma_{m(t)}$ to eliminate the denominators:

$$Y_t - \mu_{m(t)} = \sum_{i=1}^p \phi_i \left(\frac{\sigma_{m(t)}}{\sigma_{m(t-i)}} (Y_{t-i} - \mu_{m(t-i)}) \right) + \sigma_{m(t)} \varepsilon_t$$

Rearranging the equation, we get:

$$Y_t = \sum_{i=1}^p \left(\phi_i \frac{\sigma_{m(t)}}{\sigma_{m(t-i)}} \right) Y_{t-i} - \sum_{i=1}^p \left(\phi_i \frac{\sigma_{m(t)}}{\sigma_{m(t-i)}} \mu_{m(t-i)} \right) + \mu_{m(t)} + \sigma_{m(t)} \varepsilon_t$$

We can now define the new coefficients ψ_i and the new noise term η_t as follows:

$$\begin{aligned} \psi_i &= \phi_i \frac{\sigma_{m(t)}}{\sigma_{m(t-i)}} \\ \eta_t &= - \sum_{i=1}^p \left(\phi_i \frac{\sigma_{m(t)}}{\sigma_{m(t-i)}} \mu_{m(t-i)} \right) + \mu_{m(t)} + \sigma_{m(t)} \varepsilon_t \end{aligned}$$

Thus, the PAR(p) model can be rewritten in terms of the actual inflow values Y_t as:

$$Y_t = \sum_{i=1}^p \psi_i Y_{t-i} + \eta_t$$

Of course, implementing this in an efficient way requires careful handling of the time-varying coefficients ψ_i and the noise term η_t , as they depend on the month corresponding to time t and the lagged times $t - i$. However, this formulation allows us to directly incorporate the PAR dynamics into the LP model using only the actual inflow values Y_t . If ε_t is generated from sampling a normal distribution, then all the samples will have to be corrected using the above equation for η_t . All of this can be implemented in the data preprocessing step before feeding the data into the LP model. Of course, most of this can be actually pre-computed from the input data, and the LP preprocessing can be reduced to a very simple multiply and add by the new noise term.